Fraud Monitoring Dashboard: Product Requirements Document

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Abstract

Fraud continues to rise in eCommerce and digital payments, yet existing fraud detection systems often fail to provide analysts with timely, transparent, and actionable insights. This Product Requirements Document (PRD) outlines the development of a web-based Fraud Monitoring Dashboard designed to enhance fraud detection effectiveness, streamline analyst workflows, and deliver managerial visibility through reporting tools. The dashboard emphasizes real-time transaction scoring, responsive triage workflows, and role-based reporting. By closing operational gaps in existing solutions, the proposed system seeks to reduce fraud losses, improve analyst efficiency, and strengthen organizational trust in fraud mitigation strategies.

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1. Introduction

Context:

Fraud continues to rise in eCommerce and digital payments. While businesses have fraud detection engines, many lack a transparent, real-time interface that empowers analysts to monitor suspicious transactions and act quickly.

Vision:

Develop a web-based Fraud Monitoring Dashboard that provides analysts and managers with real-time visibility into suspicious activity, streamlined triage tools, and clear reporting.

Problem Statement:

Existing fraud detection tools are either too slow, too opaque, or too rigid, leading to missed fraud attempts and wasted analyst time. A tailored dashboard will close this gap.

Goals:

- Enable fraud analysts to detect and act on suspicious transactions faster.

- Reduce false positives and false negatives through improved visibility.

- Provide fraud managers with insights into fraud patterns and case resolution trends.

Link to Strategy:

Supports the broader organizational goal of reducing fraud losses, increasing trust in the platform, and improving operational efficiency for fraud teams.

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2. Objectives & Success Metrics

- **Detection Effectiveness:** Identify at least 80% of high-risk fraud attempts before fulfillment.

- **Efficiency:** Reduce average triage time per case from 12 minutes to under 5 minutes.

- **Accuracy:** Reduce false positives by 20% within 6 months.

- **Adoption:** 90% of fraud analysts using the dashboard daily within 3 months of launch.

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3. Scope

In-Scope Features:

- Transaction ingestion via API or batch upload.

- Real-time fraud scoring and flagging.

- Dashboard UI with search, filters, and alerts.

- Case triage workflow (assign, resolve, escalate).

- Role-based reporting for managers.

Out-of-Scope Features (future work):

- Direct payment processor integrations.

- Automated chargeback dispute management.

- Full machine learning model training (initial version will use rules + basic scoring).

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4. User Stories & Use Cases

User Stories:

- As a fraud analyst, I want to see flagged transactions in real time so I can act before fulfillment.

- As a fraud analyst, I want to filter cases by score, payment method, and region so I can prioritize effectively.

- As a fraud manager, I want trend reports so I can evaluate fraud patterns and team performance.

- As an engineer, I want clear API documentation so I can integrate transaction feeds.

Edge Cases & Constraints:

- Transactions with missing or malformed fields must not crash the system.

- System must support at least 10,000 transactions/day in MVP.

- Analysts may have varying screen sizes; UI should adapt responsively.

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5. Functional Requirements

- **Must Have:**

  - Real-time ingestion of transactions.

  - Fraud scoring engine with thresholds.

  - Dashboard with alerts, filters, and case detail views.

  - Case management workflow (assign/resolve).

- **Should Have:**

  - Export of case data to CSV.
  - Manager reporting dashboard.

- **Could Have:**
  - Notification system (email/Slack alerts).
  - Customizable scoring thresholds.

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6. Non-Functional Requirements
- **Performance:** Dashboard pages load in under 2 seconds.
- **Scalability:** Support up to 100k transactions/day within 12 months.
- **Accessibility:** Compliant with WCAG 2.1 AA.
- **Security:** Enforce role-based authentication and encrypted data storage.
- **Compliance:** Ensure GDPR compliance for handling user data.

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7. Dependencies & Risks
Dependencies:
- Cloud hosting (AWS/Azure/GCP).
- External fraud scoring library or rules engine.
- CI/CD pipeline for deployment.

Risks & Mitigation:
- **Risk:** Delayed data feeds reduce dashboard usefulness.
  - *Mitigation:* Implement buffer queues with retry logic.
- **Risk:** Analysts may resist adoption due to learning curve.
  - *Mitigation:* Provide onboarding materials and training.

- **Risk:** Scaling issues with data ingestion.
  - *Mitigation:* Start with throttled MVP, add load testing before scaling.

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8. Acceptance Criteria
- A transaction flagged by the fraud engine appears in the dashboard within 5 seconds of ingestion.
- Analysts can filter cases by at least 3 attributes (e.g., score, payment type, region).
- Managers can generate weekly fraud reports from the dashboard.
- Case status (open, resolved, escalated) is tracked and updated correctly.
- Authentication system prevents unauthorized access to case data.

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9. Distinction from Related Documents
- **PRD (this document):** Defines what the product must do and why.
- **PDR (Product Design Record):** Will specify UX, wireframes, and architecture choices.
- **Technical Specification:** Will define implementation details at the code and API level.

Figure 1
System Flow for Fraud Monitoring Dashboard

```mermaid
flowchart TD
  A[User Checkout] --> B[Front-End API Layer]
  B --> C[Transaction Database]
  C --> D[Fraud Scoring Engine (AI/Rules)]
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B --> E[Fraud Dashboard UI]

D --> E

E --> F[Fraud Analyst Review]

Caption: End-to-end system flow showing data ingestion, scoring, and analyst review processes.

References

- Layman, B. *Understand Django*. Chapters 3–4 (*Views On Views*, *Templates For User Interfaces*).

- [Any additional course or technical references you cited].

AI Use Disclosure

This document was created with the assistance of **ChatGPT (OpenAI GPT-5)**.

- **Prompts Used:**

 - “please provide a PRD related to my fraud dashboard”

 - “make professional submission with abstract, figures, references, explicit AI disclosure”

- **Outputs Accepted:** PRD structure, abstract draft, figure sketch.

- **Outputs Rejected/Revised:** Original AI draft was restructured for clarity, with academic formatting, captions, and citations added by the author.